CAPSTONE 31 — Live Demo Guide & Execution Log

Step-by-Step Terminal Commands & Verbatim Captured Output

1. Live Terminal Demonstration Script (Hands-On Repo, Hook & CI Demo)

This document provides the complete step-by-step live demonstration for **CAPSTONE 31 — DevOps and Git Workflow Primer with CI Basics**. Execute the commands below inside `04_Sample_Repo/` to demonstrate branch creation, automated commit-message and secret scanning, static code linting, and unit testing.

Step	Terminal Command	Expected Output & Technical Explanation
1. Verify Repository Structure	`cd 04_Sample_Repo && ls -la .github/ .github/workflows/`	Displays `pull_request_template.md`, `branch_protection_rules.json`, and `ci.yml` (GitHub Python CI Docs).
2. Install Automated Hook	`bash scripts/install_hooks.sh`	Copies `scripts/commit-msg` into `.git/hooks/commit-msg` and sets executable permissions (`chmod +x`).
3. Create Feature Branch	`git switch -c feat/sprint-velocity-validation`	Creates an isolated topic branch per GitHub Flow without affecting `main`.
4. Test Invalid Commit Rejection	`git commit -m "fixed bug in analytics"`	The `commit-msg` hook exits with status code 1 and prints `[COMMIT REJECTED]` with Conventional Commits v1.0.0 examples.
5. Test Valid Commit Acceptance	`git commit -m "feat(analytics): add velocity validation"`	Hook verifies the header regex and scans `git diff --cached` for leaked secrets, printing `[COMMIT VERIFIED]`.
6. Run Linter & Unit Tests	`python3 scripts/lint_check.py && python3 -m unittest discover -s tests -v`	Passes AST/PEP 8 static analysis (0 errors) and executes all 5 unit tests in `tests/test_analytics.py`.

2. Verbatim Captured Terminal Execution Log

The trace below was captured directly from executing the live demo inside the sample repository:

```
$ git init -b main
Initialized empty Git repository in ./04_Sample_Repo_and_CI_Workflow/.git/

$ git config user.name 'CSE Capstone Team'


$ git config user.email 'cse-team@university.edu'
```

```
$ bash scripts/install_hooks.sh
Installed Conventional Commits + Secret-Scan hook at .git/hooks/commit-msg

$ git add .


$ git commit -m 'feat(core): initialize sample repository with CI and PR template'
[main (root-commit) 9005be6] feat(core): initialize sample repository with CI and PR
template
 12 files changed, 376 insertions(+)
 create mode 100644 .github/branch_protection_rules.json
 create mode 100644 .github/pull_request_template.md
 create mode 100644 .github/workflows/ci.yml
 create mode 100644 .gitignore
 create mode 100644 README.md
 create mode 100644 requirements.txt
 create mode 100755 scripts/commit-msg
 create mode 100755 scripts/install_hooks.sh
 create mode 100755 scripts/lint_check.py
 create mode 100644 src/__init__.py
 create mode 100644 src/analytics.py
 create mode 100644 tests/test_analytics.py
[COMMIT VERIFIED] Conventional commit header & secret scan passed: 'feat(core): initialize
sample repository with CI and PR template'

$ git switch -c feat/sprint-velocity-validation
Switched to a new branch 'feat/sprint-velocity-validation'

$ python3 scripts/commit-msg ./bad_commit_msg.txt
[COMMIT REJECTED] Invalid commit message header:
  'fixed bug in analytics'

Required format (Conventional Commits v1.0.0):
  <type>(<optional-scope>): <short imperative summary (4-72 chars)>
Allowed types: feat, fix, docs, style, refactor, perf, test, build, ci, chore, revert
Examples:
  feat(analytics): add credit-weighted GPA calculator
  fix(ci): pin actions/setup-python to v5

$ python3 scripts/commit-msg ./good_commit_msg.txt
[COMMIT VERIFIED] Conventional commit header & secret scan passed: 'feat(analytics): add
sprint velocity summary validation'

$ python3 scripts/lint_check.py
[LINT PASSED] 0 errors across 3 Python files (AST syntax, <=100 cols, docstrings).

$ python3 -m unittest discover -s tests -v
test_calculate_weighted_gpa_invalid_grade
(test_analytics.TestAnalytics.test_calculate_weighted_gpa_invalid_grade) ... ok
test_calculate_weighted_gpa_mismatched_lengths
(test_analytics.TestAnalytics.test_calculate_weighted_gpa_mismatched_lengths) ... ok
test_calculate_weighted_gpa_standard
(test_analytics.TestAnalytics.test_calculate_weighted_gpa_standard) ... ok
test_summarize_sprint_velocity_negative_raises
(test_analytics.TestAnalytics.test_summarize_sprint_velocity_negative_raises) ... ok
test_summarize_sprint_velocity_standard
```

```
(test_analytics.TestAnalytics.test_summarize_sprint_velocity_standard) ... ok

----------------------------------------------------------------------
Ran 5 tests in 0.000s

OK

$ git switch main
Switched to branch 'main'

$ git status -sb
## main

$ git log --oneline --graph --all
* 9005be6 feat(core): initialize sample repository with CI and PR template
```